Lab Solution: Databases and Tables on Databricks

Setting the default catalog

Run the cell below to set the current catalog to **hive_metastore**

```
4
USE CATALOG hive_metastore
```

Q1 - Creating managed table

In the default database, create a managed table named **movies_managed** that has the following schema:

Column Name	Column Type
title	STRING
category	STRING
length	FLOAT
release_date	DATE

```
6
-- Answer
CREATE TABLE movies_managed
  (title STRING, category STRING, length INT, release_date DATE);
```

Review the extended metadata information of the table, and verify that:

1. The table type is Managed
2. The table is located under the default hive directory

```
8
DESCRIBE EXTENDED movies_managed
```

Q2 - Creating external table

In the default database, create an external Delta table named **actors_external**, and located under the directory: **dbfs:/mnt/demo/actors_external**

The schema for the table:

Column Name	Column Type
actor_id	INT
name	STRING

```
10
-- Answer
CREATE OR REPLACE TABLE actors_external
  (actor_id INT, name STRING, nationality STRING)
LOCATION 'dbfs:/mnt/demo/actors_external';
```

Q4- Checking table metadata

Review the extended metadata information of the table, and verify that:

1. The table type is External
2. The table is located under the directory: **dbfs:/mnt/demo/actors_external**

```
12
DESCRIBE EXTENDED actors_external
```

Q3- Dropping manged table

Drop the manged table **movies_managed**

```
14
-- Answer
DROP TABLE movies_managed
```

Check that the directory of the managed table has been deleted

```
16
%fs ls 'dbfs:/user/hive/warehouse/movies_managed'
```

Q4- Drop external table

Drop the external table **actors_external**

```
18
DROP TABLE actors_external
```

Check that the directory of the external table has **Not** been deleted

```
20
%fs ls 'dbfs:/mnt/demo/actors_external'
```

Q5- Creating new schema

Create a new schema named **db_cinema**

```
22
```

```
-- Answer
CREATE SCHEMA db_cinema
```

Review the extended metadata information of the database, and verify that the database is located under the default hive directory.

Note that the database folder has the extenstion **.db**

```
24
DESCRIBE DATABASE EXTENDED db_cinema
```

Use the new schema to create the below **movies** table

```
26
-- Answer
USE db_cinema;

CREATE TABLE movies
  (title STRING, category STRING, length INT, release_date DATE);
```

Q6- Creating new schema in custom location

Create a new schema named **cinema_custom** in the directory: **dbfs:/Shared/schemas/cinema_custom.db**

```
28
-- Answer
CREATE SCHEMA cinema_custom
LOCATION 'dbfs:/Shared/schemas/cinema_custom.db'
```

Use the new schema to create the below **movies** table

```
30
USE cinema_custom;

CREATE TABLE movies
  (title STRING, category STRING, length INT, release_date DATE);
```

Finally, review the extended metadata information of the table **movies**, and verify that:

1. The table type is Managed
2. The table is located in the new database defined under the custom location

```
32
DESCRIBE EXTENDED movies
```

```
33
```